

range Qualifier Syntax

```
integer-type [ range min..max ]           // explicit constraint (inclusive)
integer-type [ range 8-bits ]             // width constraint
integer-type [ range 16-bits ]
integer-type [ range 32-bits ]
integer-type [ range 64-bits ]
```

The **range** qualifier MAY be used to constrain the range of values for a numeric type such as **SIGNED INTEGER**, **UNSIGNED INTEGER**, **FLOAT32**, or **FLOAT64**. Two forms are supported: *explicit constraints* and *width constraints*. Only one form MAY be applied to a given type.

An *explicit constraint* gives specific minimum and maximum (inclusive) values for the type. These MAY be any value that is legal for the underlying type.

A *width constraint* constrains the value to fit within a specific number of bytes. Any of the width constraints (8-bits, 16-bits, 32-bits or 64-bits) MAY be applied to **SIGNED INTEGER** and **UNSIGNED INTEGER** types, where 8-bits, 16-bits, 32-bits and 64-bits constraints correspond to 1-octet, 2-octet, 4-octet and 8-octet **element type** respectively; only 32-bits constraint MAY be applied to **FLOAT32** type and only 64-bits constraint MAY be applied to **FLOAT64** type.

Note that a width constraint **range** qualifier does not obligate an encoder to always encode the specified number of bits. Per the TLV encoding rules, senders are always free to encode integer and floating point values in any encoding size, bigger or smaller, that will accommodate the value.

range Qualifier Example

```
system-status-event => STRUCTURE
{
    timestamp [1]           : UNSIGNED INTEGER [ range 32-bits ],
    num-processes [2]       : UNSIGNED INTEGER [ range 16-bits ],
    percent-busy [3]        : UNSIGNED INTEGER [ range 0..100 ],
}
```

B.5.8. tag*tag Qualifier Syntax*

```
identifier [ tag-num ]           // context-specific tag
identifier [ protocol-id:tag-num ] // protocol-specific tag
identifier [ protocol-name:tag-num ] // protocol-specific tag
identifier [ *:tag-num ]         // protocol-specific tag (cur. protocol)
identifier [ anonymous ]         // no tag
```

The **tag** qualifier is allowed on type names, field names within a **STRUCTURE** (**STRUCTURE Fields**) or **FIELD GROUP**, item names within a **LIST** (**LIST Item Tags**), alternate names within a **CHOICE OF** (**CHOICE**